Chapter 1

Experimental Setup and Protocol

Chapter **??** described Strategies 0–3 (the uncompensated baseline, pure INDI, geometric+NN and FBL+NN), each treated as its own controller. This chapter describes the platform, the scenarios, the measurement chain and the protocol, and in each case the reasoning is the same: the design exists to remove alternative explanations for an observed difference.

Section 1.7 states the threats this design is built against, and which of them it does not eliminate.

Campaign status (September 2026). Two-robot flights with OptiTrack and onboard logging at 500 Hz have been executed on the brushless pair **cf5** (study vehicle) and **cf_second** (top). Micro-SD merges with per-vehicle role alignment replaced radio-only windows for thesis metrics. On **2026-09-21**, eight verified geometric merges were archived (five C.1 training-eligible A1/A3 cells plus three A8 QA shakedowns; manifest `c1_2026_09_21_merged/manifest_2026_09_21_c1.json`), with dashboards and phase metrics produced by the standard analysis suite. Indicative Strategy 0 vs 1 comparisons exist from earlier sessions; the systematic comparison (C.4) still requires repeated flights per controller and scenario. Residual-model training data collection (C.1) *has begun* under **geometric control on cf5** so measured a_{res} is not cancelled by full INDI, but the planned dz/speed grid is only partly filled (see `docs/25` coverage table).

1.1 Platform

The vehicles are Crazyflie 2.1 brushless quadrotors, nominally 41 g and 9 cm rotor-to-rotor. All vehicles in a team are the same platform: the gain block is shared across robots, so a mixed team would fly at least one vehicle on the wrong mass, thrust constant and inertia, which would appear in the logs as a residual and be indistinguishable from an interaction force (Section **??**).

Two vehicles are the minimum for the protocol. A third is required for the three-robot scenarios, which cannot be run without it.

Each vehicle carries two additions that are not optional for this work:

A rotor-speed source , optical RPM deck or DShot telemetry. Without it the applied thrust cannot be reconstructed, so a_{res} is unavailable and *reads identically zero* rather than failing loudly. Strategy 1 additionally cannot run at all. This is the single most consequential hardware prerequisite in the setup.

A micro SD deck , the recording medium for the dataset, for the reasons in Section 1.4.

State estimation uses the on-board extended Kalman filter, fed with external pose from a motion-capture system at 100 Hz. The control loop runs at 500 Hz onboard.

1.2 Flight volume

The tracked volume measures approximately 2 m × 4 m horizontally and extends to 1.7 m in height. Two distinct limits are enforced and should not be conflated: the *tracked volume*, outside which the motion-capture system cannot see the vehicles, and an *operational floor* of 0.30 m, chosen so that ground effect does not contaminate the measurement of inter-vehicle interaction.

The floor is lowered to 0.10 m for one scenario only, the near-ground pass of Section 1.3, whose purpose is to measure ground effect in isolation so that it can be separated from downwash rather than confounded with it.

The volume is a binding constraint rather than a formality. Several scenarios sit within approximately 10 cm of a wall at their extremes, and the scenario definitions are automatically centred and, where necessary, rotated to fit. That a configuration fits is checked before flight, not discovered during it.

The volume figures are corroborated from two independent statements but have not been tape-measured. Measure before the first session and update this section (docs/11_Hardware_Readiness_Checklist.md).

1.3 The formation library

Sixteen scenarios, fixed before any data collection and not modified afterwards. Freezing them is what allows every strategy to meet identical excitation; a library that evolved during the campaign would make later strategies incomparable with earlier ones.

The scenarios are parameterised rather than hard-coded, separations, offsets and speeds are arguments, so that a separation sweep is the same scenario at different parameters rather than a different scenario. Each is compiled to a per-vehicle polynomial trajectory and uploaded through the standard trajectory interface, so that the same command produces the same motion in simulation and on hardware.

1.3.1 The null scenarios

C1–C3, the coplanar controls in which the vehicles fly side by side at equal height, are the most important scenarios in the library and the least interesting to watch. They place the vehicles in configurations where no significant inter-vehicle interaction should exist, and they exist because of the property established in Section **??**: the residual is a catch-all, non-zero even in isolated flight, so a residual measured in formation cannot be attributed to the interaction without a reference.

They serve two purposes. They bound the measurement floor, so that a reported improvement can be distinguished from noise. And they detect model error: a residual that appears when no neighbour is in the wash is a mis-identified mass or thrust constant, not an interaction, and must be corrected before the campaign proceeds rather than explained afterwards.

1.3.2 Regimes, and why the library spans them

The scenarios are not a list of poses but a deliberate spread across the interaction regimes that Research Question 2 distinguishes. A1, the two-robot vertical hover stack, and the coplanar controls C1–C3 are static; A2 and A3 are quasi-static; A4 and A5 sweep lateral offset; A7 sweeps separation continuously; A8 and B3 present the disturbance as a transient with a known arrival time. A model fitted on one regime and evaluated on another is what makes "outside the training distribution" a measurable condition rather than a caveat.

> **Requirement for data collection: the set must excite lateral relative motion.** A purely vertical scenario holds the lateral components of the relative state constant, so a model trained only on such data has seen no variation in half of its six inputs and is extrapolating the first time a formation moves sideways. **The collection set must therefore include A4, A5 and A7 — not only the vertical stacks A1, A2 and A3.** This is not a preference about coverage; a set without them produces a model that cannot be evaluated on the scenarios it will meet.

This was established in simulation before any flight: a dry run collecting only a vertical stack produced training data in which two of the six network inputs never varied, which the training pipeline reports as a degenerate normalisation statistic and which no amount of training can repair.

1.4 Measurement and logging

1.4.1 Why the dataset is recorded onboard

Radio telemetry saturates at approximately two vehicles: each drone transmits on the order of 800 packets per second at full rate against a per-dongle ceiling near 1000. Recording the dataset over radio would therefore couple the sampling rate to the team size, which is unacceptable for a study whose subject is what happens as vehicles are added.

The micro SD deck records 35 variables at 500 Hz on each vehicle independently of radio and of team size. Radio logging is retained for live monitoring and for single-vehicle tuning, where the bandwidth exists.

1.4.2 What is recorded

Position and velocity estimates; attitude; gyro and accelerometer; the measured residual a_{res}; the INDI torque and filtered angular acceleration; the network prediction $\hat{a}_{\text{res}}$ and its saturation flag; the commanded setpoint; and the four motor commands.

Two of these deserve comment. The measured residual is recorded *in every controller mode*, including those that make no use of it, because the training data is collected under the uncompensated baseline. And the network prediction is recorded *whether it is enabled or not*, because comparing prediction against measurement is how a learned model is evaluated, and that comparison is only meaningful on a flight where the prediction is not also changing the vehicle's behaviour.

The logging configuration is fixed before data collection and shared by the collection and evaluation flights. A configuration change between the two would silently make the two halves of the dataset incomparable.

1.4.3 Time alignment across vehicles

Each vehicle timestamps its samples against its own boot, so the logs share no clock. Recording is started by a single broadcast, which aligns the vehicles to within the broadcast jitter plus one logging period. Residual error comes from propagation, task wake phase and crystal drift between vehicles; the last is a rate error and accumulates over a flight.

The merge tool therefore *measures* the offset by cross-correlating each vehicle's onboard trace against a radio trace sharing a common clock, and reports offset and drift rather than assuming them. The predicted alignment is a few milliseconds; that figure is to be confirmed against real logs on the first two-vehicle flight.

1.5 Protocol

1.5.1 Hardware gate

No data collected before the following sequence passes is used. The gate is not an experiment; it is the check that nothing is broken, so that what is measured afterwards means something.

1. **Validate.** Single vehicle, geometric first and then INDI, on the trajectory ladder. Confirm the gains read back as configured.
2. **Tune, then freeze.** Any remaining attitude tuning happens here and only here. The gains are then frozen for the remainder of the campaign.
3. **Two vehicles at large separation.** Confirm a_{res} is non-zero and correctly signed; confirm the measured time alignment.

The order carries the reasoning. Tuning before validating means tuning against an unknown fault. Flying two vehicles before the freeze means the gains move underneath the dataset.

Stage 1 cannot confirm that the compensation terms are *correct*, only that they are not catastrophic: with no neighbour present $a_{\text{res}} \approx 0$, so a compensation term is dormant and a clean single-vehicle flight demonstrates nothing about it. Stage 3 is the first point at which the sign of the compensation is observable at all, which is why it is part of the gate rather than the first experiment.

1.5.2 Data collection

Collection flights are flown under the *uncompensated* baseline of Section **??**. This is not a convenience: a compensating controller partly cancels the disturbance, so the residual it records is the one that survived compensation, which is not the quantity a predictive model should be fitted to.

Separations are approached from large to small, and a tighter configuration is attempted only after the looser ones are clean.

The set must span both vertical and lateral excitation — A1/A2/A3 *and* A4/A5/A7 — for the reason given in Section 1.3. A collection campaign that omits the laterally excited scenarios cannot support the predictive strategies, and the omission is not recoverable without re-flying.

1.5.3 Comparison campaign

Each strategy flies the frozen library. Because the control law is selected at runtime (Section **??**), strategies are interleaved within a session rather than flown on separate days, which pairs them against the same battery state, temperature and calibration. Each condition is repeated to allow a distribution rather than a single number to be reported.

Each strategy is compared under identical trajectories, formations, hardware and separations, with its own gains frozen after the hardware gate, not identical control structure — this holds uniformly across all of them, not just a subset.

1.5.4 Metrics

Three axes, corresponding to the research questions.

Tracking error. Position RMSE, total and per axis, and the maximum deviation from the commanded trajectory. Alongside these, the attitude tracking error, reported as the RMSE and peak of the geometric attitude error e_R. A compensator that improves position by working the attitude loop harder has not improved tracking so much as moved the cost, and only the attitude figure makes that visible. The logging configuration described in Section 1.4.2 records attitude but not e_R itself; that signal will be added before the comparison campaign.

Residual rejection. The error in the *separation between vehicles*, measured against the commanded separation. Downwash acts as a bias, the lower vehicle sags, closing the gap, so the mean error is reported as the primary quantity and the spread as secondary. A compensator that works moves the mean back toward the commanded value. Reporting only the spread would hide precisely the effect being studied.

Alongside it, the magnitude of the measured residual that remains after compensation, and for every strategy carrying a learned model, the prediction against the measurement.

Robustness and cost. Control effort; behaviour as separation tightens; abort and failure rate; onboard computation per control step; sensing prerequisites; and the quantity of training data required to reach a given prediction accuracy.

1.6 The role of simulation

A software-in-the-loop simulator runs the same controller source that flies, with a published downwash model providing the interaction force. It was used throughout development, and its role in this thesis is deliberately limited.

Simulation is used to disprove implementations, never to validate a method. It established that every scenario in the library executes and produces the commanded geometry under both controllers, and it exposed two defects in flight code that single-vehicle testing could not have found. It cannot support a claim about how well a method works, for a reason specific to this subject: the interaction force in simulation is produced by a learned model of the same family as the one being fitted, and learning a network's output with another network is a substantially easier problem than learning real air.

No result in the two-robot results chapter or the multi-robot results chapter rests on simulation.

1.7 Threats to validity

Splitting the training and test data by contiguous flight segment rather than by sample deserves a note, because the obvious alternative is badly wrong here. At 500 Hz consecutive samples are nearly identical, so a random per-sample split places a near-duplicate of almost every test point in the training set and reports a validation error that means nothing.

Two threats are not mitigated and are stated as limitations rather than managed. The results concern a single airframe; quantities are normalised where a normalisation is defensible, but generalisation to other platforms is argued rather than shown. And battery discharge changes the thrust constant slightly within a flight, which appears as a slow drift in the measured residual; the effect is bounded by the null scenarios but not removed.

1.8 Summary

Two or three identical brushless quadrotors, each carrying a rotor-speed source and an SD card, flying sixteen frozen scenarios inside a tracked volume of roughly 2 m × 4 m × 1.7 m, with 35 variables recorded onboard at 500 Hz independently of team size. Each strategy is tuned on the hardware gate. Those gains are frozen before the comparison and training-collection flights; strategies are selected at runtime and interleaved within sessions; and three null scenarios establish what a measurement of zero looks like under the same pipeline.

Continue revising to past tense for each scenario and controller row as C.4 repeats land. Record any deviation from a per-strategy gain block, frozen after the hardware gate, or from scenario parameters in this chapter.

Table 1.1: The formation library. Scenarios marked † carry elevated risk and are gated behind an explicit acknowledgement; those marked ∗ are null conditions in which no inter-vehicle interaction is expected.

	Scenario	Purpose
Two vehicles		
A1	Vertical stack, hover	Baseline. Nothing moves, so any relative displacement is interaction rather than tracking error
A2	Vertical stack, tracking	Constant separation while both vehicles move
A3	Static top	Fixed wash source, lower vehicle sweeps through: captures the transition, not only the level
A4	Offset stack	Partial overlap; asymmetric and hardest to model
A5	Reverse-circle stack	Lateral offset swept twice per lap at constant height and speed
A6	Extreme stack †	Strongest interaction, smallest error budget
A7	Dynamic merge †	One continuous sweep of the whole separation range: gives a model the gradient, not only levels
A8	Vertical swap	Wash arrives as a transient with a known arrival time; hardest case for a feedforward model
Three vehicles		
B1	I-stack	Lowest vehicle in the combined wash of two
B2	V-stack	Asymmetric aggregation, which B1 cannot produce
B3	Vertical swap	Three crossings at three height differences in one flight
Controls and coverage		
C1	Side-by-side coplanar ∗	A residual here, and not in A1, is not downwash
C2	Leader–follower line ∗	Wake rather than wash
C3	Equilateral triangle ∗	Coplanar three-body control
C4	Docking approach †	Simultaneous lateral and vertical closure
C5	Near-ground pass, one vehicle	Ground effect alone, separable from downwash

Table 1.2: Threats to validity and the design decisions taken against them.

Threat	Mitigation	Residual risk
Tuning confound	Each strategy tuned on the hardware gate; those gains frozen before comparison flights; runtime selection allows interleaving	Per-strategy tuning effort is in scope for RQ4 (§**??**)
Measurement floor	Null scenarios C1–C3 bound it under the same pipeline	—
Model error read as interaction	Null scenarios detect it; airframe constants identified once and frozen	Battery discharge within a flight is not compensated
Training/test leakage	Held-out scenarios; split by contiguous flight segment, never by sample	—
Session effects	Strategies interleaved within a session; repeated conditions	—
Silent loss of the residual	$a_{\text{res}} \neq 0$ verified at the start of every session	—
Radio vs onboard clock	uSD logs at 500 Hz; analysis uses merged uSD and firmware `ctrltarget.*` where available	Residual-time alignment across vehicles depends on merge quality; `merge_usd_logs.py` aborts above 15 cm RMS role check
Mocap identity / marker swap	Pre-takeoff EKF-vs-mocap gate; role verification from merge	Residual risk if a session bypasses the gate
Simulation-to-hardware gap	No claim rests on simulation	—
Single airframe	—	**Not mitigated.** Generalisation beyond this platform is discussed, not demonstrated

Bibliography